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ORIGINAL INVESTIGATION

Effects of Far-Infrared Radiation-Lamp Therapy on Recovery From Simulated Soccer Match Running Activities in Elite Soccer Players

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Purpose: The authors investigated whether far-infrared radiation (FIR) lamp therapy would reduce muscle damage and enhance recovery from multiple soccer-match-related running activities. ***Methods:*** Twenty-four elite female soccer players (20–24 y) were assigned into a FIR or a sham treatment group (n = 12/group). They performed a daily 90-minute Loughborough Intermittent Shuttle Test (LIST) for 6 consecutive days. Maximal voluntary contraction torque of the knee extensors (KEs) and flexors, muscle soreness, plasma creatine kinase activity, countermovement jump, and several other performance measures (eg, 30-m dash, Yo-Yo Intermittent Recovery Test Level 1) were taken before the first LIST, 1 hour after each LIST, and 24, 48, 72, 96, and 120 hours after the last LIST. All participants received a 30-minute FIR or sham treatment on KEs and knee flexors, respectively, at 2 hour after each LIST and 25, 49, 73, and 97 hours after the last LIST. ***Results:*** All measures changed significantly ($P < .05$) at 1 hour after the first LIST without difference ($P > .05$) between groups. Maximal voluntary contraction torque (eg, the largest decrease of KE for FIR: 13% [4%], sham: 25% [5%]), countermovement jump height (4% [3%] vs 14% [4%]), and other performance measures (eg, Yo-Yo Intermittent Recovery Test: 11% [5%] vs 26% [5%]) decreased less, and peak muscle soreness (eg, KE: 26 [9] vs 51 [18] mm) and plasma creatine kinase activity (172 [32] vs 1289 [610] IU/L) were smaller for the FIR than for the sham group ($P < .05$), and they returned to the baseline earlier ($P < .05$) for the FIR group. ***Conclusions:*** These results suggest that the FIR therapy provided potent effects on reducing accumulated muscle damage and enhancing recovery.

Keywords: 90-minute Loughborough Intermittent Shuttle Test, muscle damage, countermovement jump, 30-m dash, Yo-Yo Intermittent Recovery Test Level 1

Muscle damage is induced by a soccer match, resulting in symptoms such as decreased muscle function, delayed-onset muscle soreness (DOMS), and impaired athletic performance.[1,2] These symptoms could last for several days after a soccer match and affect training, as well as the next match performance.[2–4] Chou et al[2] investigated muscle damage after a single and multiple 90-minute Loughborough Intermittent Shuttle Test (LIST) that were performed by elite female soccer players one day or every day for 3 or 6 consecutive days. They reported that the symptoms of muscle damage lasted for 3 to 5 days after a single match, the total distance covered during the LIST was decreased in the second LIST onward, and decreases in muscle strength and other performance measures did not recover to the baseline even at 5 days after the sixth LIST. Thus, strategies to enhance recovery from fatigue and muscle damage are necessary.

To enhance the recovery from muscle damage in relation to soccer, a variety of interventions including a cold water immersion, foam rolling, compression garments, electrical stimulation, and far-infrared radiation (FIR) therapy have been investigated.[5–13] Although some effects of the interventions on recovery were observed, the effects appear to be inconclusive. To the best of our knowledge, 3 studies have examined the effect of a FIR therapy on muscle damage relating to soccer or soccer players.[11–13] Coelho et al[11] used a FIR "pants" worn for 2 nights during sleep (~8 h each) following an official soccer match performed by male university players. They found no significant effects of the FIR intervention on countermovement (CMJ) and squat jump, 20-m sprint, DOMS, and plasma creatine kinase (CK) and lactate dehydrogenase activities. Nunes et al[12] examined the effects of a FIR pants worn during sleep daily for 2 weeks by Brazilian elite futsal players during a preseason heavy training period on subjective training strain assessed by a 10-point rate of perceived exertion scale, performance measures (squat jump, CMJ, and sprints), DOMS, and some biochemical markers. They found some effects on the training strain and serum tumor necrosis factor-α concentration but no effects on other measures. Loturco et al[13] investigated the effect of a FIR garment worn during sleep (10 h) over 3 consecutive nights following a bout of 100 drop jumps performed by elite male soccer players. They found no significant therapeutic effects on muscle damage markers. However, wearing FIR pants or garment is not only the FIR intervention.

Hausswirth et al[14] reported that a 30-minute FIR lamp therapy applied to the whole body immediately, 1 and 2 days after a 48-minute simulated trail run increased the recovery of knee extensor (KE) strength at 1 day postrunning. In a review paper, Leung[15] has stated that FIR has therapeutic effects to delay muscle fatigue, reduce chronic muscle and joint pain, increase microcirculation, and enhance metabolism. The underpinning mechanisms of FIR therapy effects are not clear yet, but previous studies suggest that FIR produce both nonthermal and thermal effects including an increase in microvascular dilation, increased blood flow, and elevated regional tissue temperature, albeit in rats.[16,17] Thus, it is interesting to investigate the effects of FIR lamp therapy on muscle damage accumulated in multiple simulated soccer matches.

Therefore, the present study tested the hypothesis that daily FIR lamp therapy for 10 days over 6 simulated LISTs performed by

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elite female soccer players would reduce accumulative fatigue and muscle damage and enhance recovery of performance parameters when compared with a sham treatment.

Methods

Participants and Study Design

Twenty-four elite female university soccer players who were winners of the 2018 Taiwan University Football Tournament provided informed consents to participate in the present study. The study was approved by the Research Ethics Committee of National Taiwan Normal University in Taiwan (approval number: 201812HM023). The study was conducted in conformity with the policy statement regarding the use of human subjects by the Declaration of Helsinki. The participants played 17 to 20 official matches during the experimental period of 40 weeks and had training sessions 5 days a week (~3 h per session). The present study was conducted during an off-season. The investigators of the present study did not have any potential conflicts of interest with the participants.

The participants were randomly placed into a FIR or a sham treatment group (n = 12/group) by stratifying them, based on the preexercise maximal voluntary isometric contraction (MVC) torque of the dominant KEs (MVC-KE toque) and the baseline average MVC-KE torque to be similar between groups. No significant ($P > .05$) differences in age (21.0 [1.4] y), height (160.8 [5.5] cm), body mass (57.8 [7.2] kg), body mass index (22.3 [2.1] kg/m^2), percentage of body fat (15.2% [4.5%]), maximal oxygen consumption (VO_2max: 54.0 [4.3] mL/kg/min), and MVC-KE torque (159.4 [15.7] N·m) were found between the FIR and sham groups.

All participants performed a 90-minute LIST every day for 6 consecutive days.[2] The participants in the FIR group received a daily FIR treatment for the bilateral KEs (30 min) and bilateral knee flexors (KF; 30 min) as shown in Figure 1, at 2 hours after each LIST, and at 25, 49, 73, and 97 hours after the last LIST, and those in the sham group received a pseudo-FIR treatment (no radiation) at the same time points shown above.

The sample size was estimated using our pilot study data for a possible effect size of 1 for the difference between the FIR and sham treatment conditions for changes in MVC-KE torque after 6 LISTs. It was estimated that at least 11 participants were necessary for each condition, with an alpha level of .05 and power ($1 - \beta$) of 0.80.[18] Considering a possible dropout and an estimation error, 12 players were recruited for each group.

Familiarization Session

In a familiarization session that was set 6 to 7 days before the first LIST, the participants experienced the measurements of muscle soreness, CMJ, submaximal, and MVC at 90° and 30° of knee flexion for KEs and KF, respectively, on an isokinetic dynamometer (Biodex System S4, Biodex Medical Systems), 30-m dash, 30-m timed hop, balance, agility *T* test, 6 × 10-m shuttle run and Yo-Yo intermittent recovery test level 1, and performed a 20-minute LIST. They also practiced the running corresponding to the velocity of 55% and 95% of their VO_2max for 5 minutes for each intensity.[2]

VO_2max Test

The VO_2max was measured in a treadmill running test.[1] The running speeds corresponding to 55% and 95% VO_2max were calculated for the LIST shown below.

The Loughborough Intermittent Shuttle Test

A 90-minute LIST consisting of 5 sets of 15-minute intermittent running and intermittent shuttle run to exhaustion was performed in an indoor hall.[19] During each LIST, the distance covered, heart rate (HR), rating of perceived exertion (RPE), and blood lactate concentration were measured and recorded. The details of the LIST were described elsewhere.[2]

FIR and Sham Treatments

Two FIR lamps were used to cover the right and left legs for both FIR and sham groups (Figure 1). The FIR group received 30 minutes of a FIR lamp treatment (wavelength: 8–14 μm, emissivity: 0.96) for the KE and KF, each using a commercially available device (EQ-A718 ITU Far-Infrared Therapy Unit, Chen Yeh Technology Health Co, Ltd) at 2 hours after each LIST, and at 25, 49, 73, and 97 hours after the sixth LIST. The FIR device was preheated for 10 minutes to ensure that the far-infrared rays reached the highest emissivity. This preparation was done before the participants came to the treatments. For the sham treatment, the

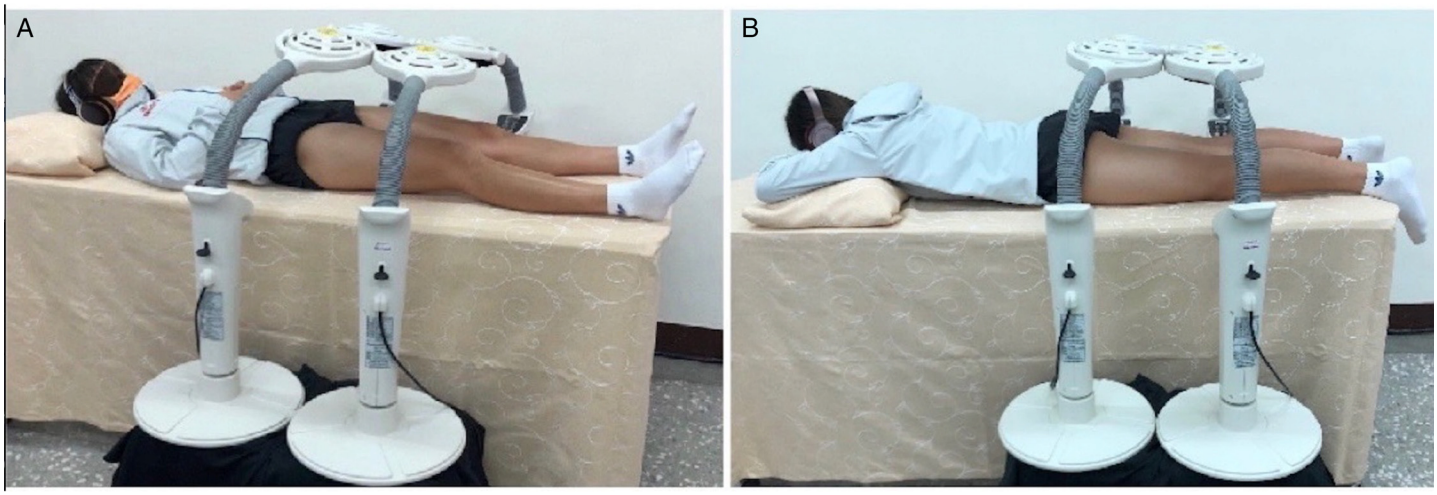

Figure 1 — Far-infrared radiation therapy setup for the bilateral knee extensors (A) and flexors (B). Two far-infrared radiation lamps were used for each leg to cover the entire knee extensors or flexors for 30 minutes a session.

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same model device without FIR radiation was used for the same time points. Each participant received a treatment for the KE first for 30 minutes followed by the KF for 30 minutes (Figure 1). The FIR irradiation surface was aligned with the muscle belly of the KEs (Figure 1A) and KF (Figure 1B), and the distance between the FIR lamp and the muscle belly was 20 cm. To eliminate potential psychological effects, each participant wore an eye mask and a headphone with music of her choice (Figure 1) to prevent any speculation for the treatment (FIR or sham). The same physical therapist carried out the FIR and sham treatments, and this was the only person who knew the condition of the treatment (FIR, sham) of each participant during the data collection period.

Muscle Damage Markers

The muscle damage markers included MVC torque of the KE and KF, CMJ height, muscle soreness of the KEs and KFs, and plasma CK activity. These markers were adopted from our previous study, and the details were described elsewhere.[2,20–22]

Performance Indices

The performance indices consisted of 30-m dash, 30-m timed hop test, balance test, agility *T* test, 6 × 10-m shuttle run, and Yo-Yo intermittent recovery test level 1. They were adopted from our previous study, and the details were described elsewhere.[2]

Statistical Analyses

Data were verified by a Shapiro–Wilk test for the normality and a Levene test for the homogeneity of variance assumption. All dependent variables before the first LIST between FIR and sham groups were compared by the independent *t* test. Changes in HR, RPE, blood lactate concentration, and total distance covered during the LIST were compared between the FIR and sham groups by a mixed design of 2-way analysis of variance (ANOVA). Changes in each dependent variable before, 1 hour after each LIST, and every 24-hour interval for 5 consecutive days after the last LIST were compared between the FIR and sham groups by a mixed design of 2-way ANOVA. When a significant interaction effect was found, a Tukey post hoc test was performed. Eta-squared values (η^2) were calculated as measures of effect size, and they were considered as ~.02, small effect; ~.13, medium effect; and >.26, large effect.[23] A significant level was set at $P < .05$. The data were presented as mean (SD).

Results

No significant differences ($P > .05$) in any of the baseline dependent variables and their changes before the first LIST existed between the FIR and sham groups (Figures 2–4). The changes in average total distance covered, HR, RPE, and post-LIST blood lactate concentration in the first LIST was also similar ($P > .05$)

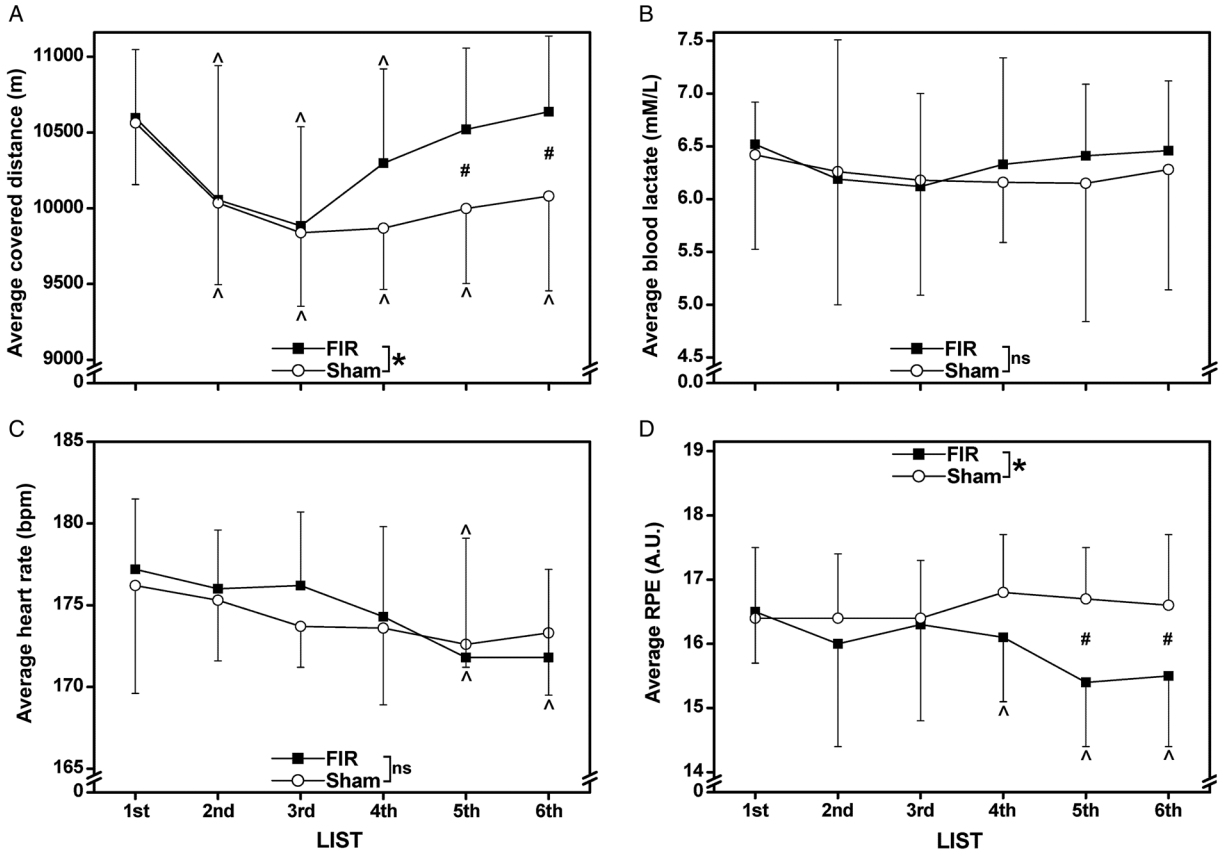

Figure 2 — Average changes (mean [SD]) in the total distance covered (A), blood lactate concentration (B), heart rate (C), and ratings of perceived exertion (RPE; D) in the first to sixth LISTs for the FIR and sham treatment groups. ANOVA indicates analysis of variance; A.U., arbitrary units; bpm, beats per minute; FIR, far-infrared radiation; LIST, Loughborough Intermittent Shuttle Test. *Significant ($P < .05$) interaction effect by mixed-design 2-way ANOVA. #Significant ($P < .05$) difference between groups for each time point based on the post hoc tests. ^Significant ($P < .05$) difference from the first bout value.

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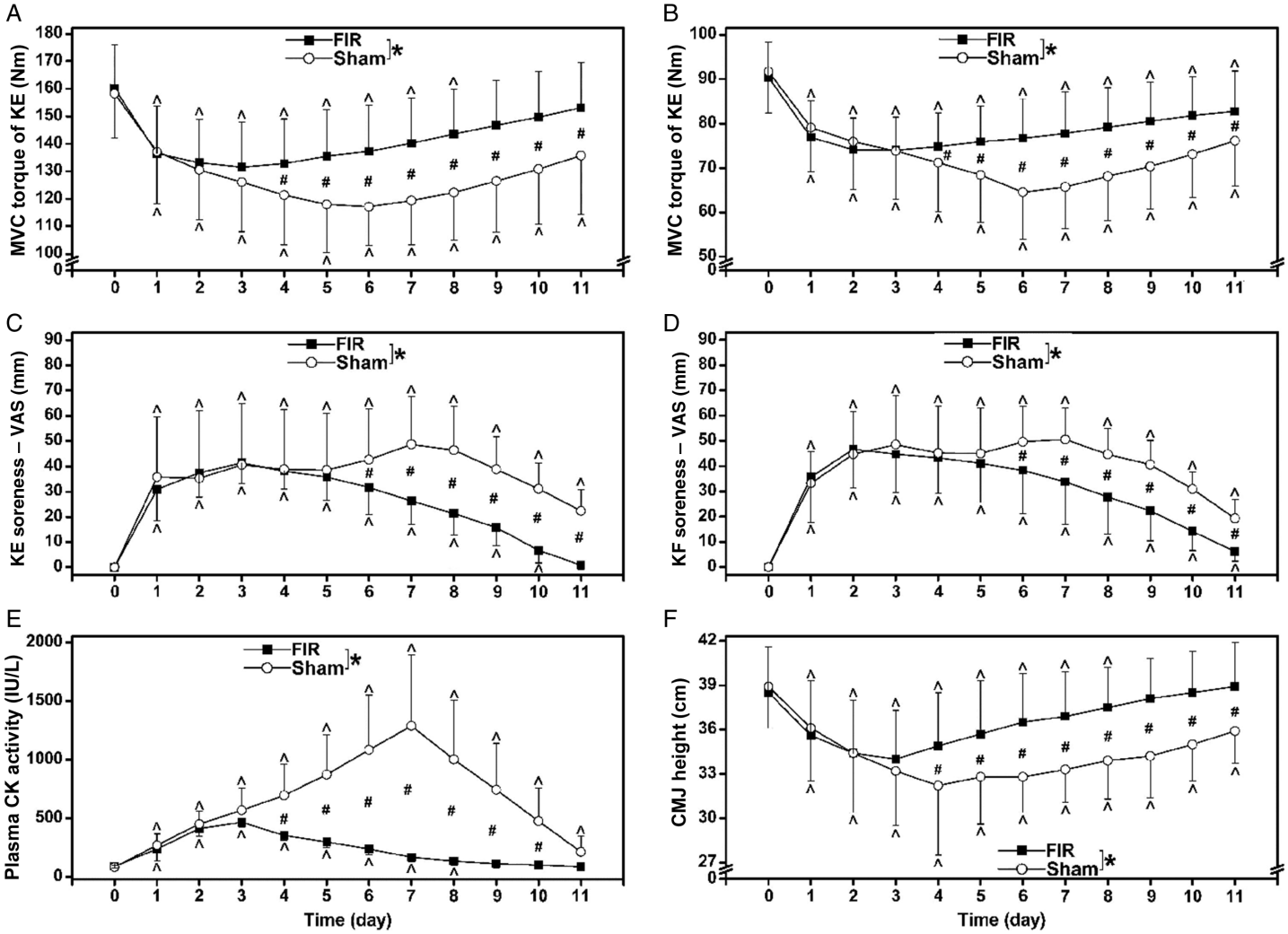

Figure 3 — Changes (mean [SD]) in MVC torque of the KEs (A) and KFs (B), muscle soreness assessed by a 100-mm VAS of the KEs (C) and KFs (D), plasma CK activity (E), and CMJ height (F) at baseline (0), 1 hour after each LIST performed 6 consecutive days in a row (1–6), and 1, 2, 3, 4, and 5 days (7–11) after the sixth LIST for the FIR and sham treatment groups. ANOVA indicates analysis of variance; CK, creatine kinase; CMJ, countermovement jump; FIR, far-infrared radiation; KE, knee extensor; KF, knee flexor; LIST, Loughborough Intermittent Shuttle Test; MVC, maximal voluntary isometric contraction; VAS, visual analog scale. *Significant ($P < .05$) difference between groups based on the interaction effect shown by the ANOVA. #Significant ($P < .05$) difference between groups for each time point based on the post hoc tests. ^Significant ($P < .05$) difference from the baseline (0) value.

between the groups. No significant ($P > .05$) differences in the average HR and post-LIST blood lactate concentration were found between the FIR and sham groups in the first to sixth LISTs (Figure 2). However, a significant interaction effect was found for the total distance covered ($P = .019$, $\eta^2 = .212$) and RPE ($P = .007$, $\eta^2 = .246$). The post hoc tests revealed that the FIR group had greater distance and smaller RPE ($P < .05$) than the sham group in the fifth and sixth LISTs.

All dependent variables of muscle damage and performance showed significant ($P < .05$) changes at 1 hour after the first LIST without significant ($P > .05$) differences between the groups (Figures 3 and 4). However, the extent of decreases in MVC-KE ($P = .002$ to $P < .001$) and MVC-KF torque ($P = .019$ to $P < .001$) as well as CMJ height ($P = .016$ to $P = .002$) at 1 hour after the fourth to sixth LIST were significantly smaller for the FIR than sham group. The recovery of MVC-KE torque (interaction effect: KE: $\eta^2 = .700$), MVC-KF torque ($\eta^2 = .788$) and CMJ height ($\eta^2 = .667$) at 1 to 5 days after the last LIST was significantly faster ($P < .01$) for the FIR than sham group (Figure 2A, 2B, and 2F), and the variables returned to the baseline faster for the FIR group. The extent of increases in the muscle soreness of KE (interaction effect; $\eta^2 = .507$; Figure 3C) and KF ($\eta^2 = .347$; Figure 3D) and plasma CK activity ($\eta^2 = .789$; Figure 3E) was significantly ($P < .001$) smaller for the FIR than sham group.

A significant interaction effect was also evident for the changes in 30-m dash ($\eta^2 = .570$; Figure 4A), 30-m timed hop ($\eta^2 = .890$; Figure 4B), balance ($\eta^2 = .804$; Figure 4C), agility *T* test ($\eta^2 = .673$; Figure 4D), 6 × 10-m shuttle run ($\eta^2 = .735$; Figure 4E), and Yo-Yo intermittent recovery test level 1 ($\eta^2 = .638$; Figure 4F). The peak changes in these performance measures were smaller ($P < .01$) for the FIR than sham group, and many of these measures returned to the baseline earlier for the FIR group.

Discussion

The main findings of the present study were as follows: (1) The distance covered in the fifth and sixth LISTs was significantly

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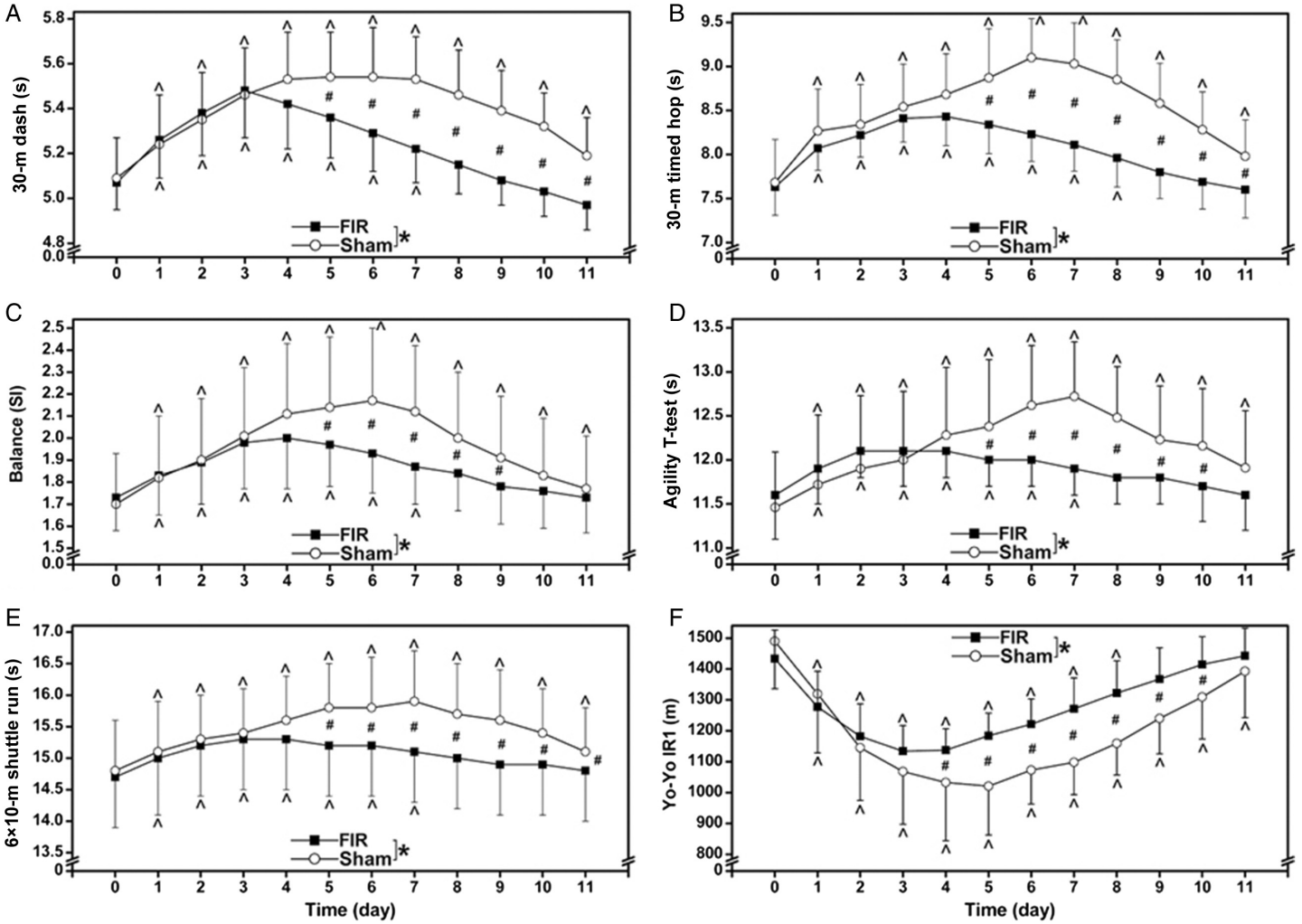

Figure 4 — Changes (mean [SD]) in 30-m dash (A), 30-m timed hop test (B), balance with eyes closed on unstable surface (C), agility *T* test (D), 6 × 10-m shuttle run (E), and Yo-Yo Intermittent Recovery Test Level 1 (F) at baseline (0), 1 hour after each LIST performed 6 consecutive days in a row (1–6) and 1, 2, 3, 4, and 5 days (7–11) after the sixth LIST for the FIR and sham treatment groups. ANOVA indicates analysis of variance; FIR, far-infrared radiation; LIST, Loughborough Intermittent Shuttle Test. *Significant ($P < .05$) difference between groups based on the interaction effect shown by the ANOVA. #Significant ($P < .05$) difference between groups for each time point based on the post hoc tests. ^Significant ($P < .05$) difference from the baseline (0) value.

longer, but their RPE was significantly lower for the FIR than sham group (Figure 2); and (2) changes in muscle damage and performance parameters were significantly smaller, and they returned to the baseline values faster for the FIR than sham group (Figures 3 and 4). These findings supported the hypothesis that a daily FIR lamp therapy would reduce accumulative fatigue and muscle damage and enhance the recovery from the multiple LISTs performed for 6 consecutive days.

As shown in Figure 2, the total distance covered and RPE during first 4 LISTs were similar between the FIR and sham groups. It should be noted that the treatment was applied at 2 hours after each LIST, and at 25, 49, 73, and 97 hours after the sixth LIST (10 times in total). The distance and RPE were significantly greater (5%–6%) and smaller (6%–8%), respectively, in the fifth and sixth LISTs for the FIR than sham group. The results shown by the sham group were similar to those shown in the previous study[2] in which 6 LISTs were performed without any treatment. Thus, it seems likely that the sham treatment group represented the effects of multiple soccer match-related running activities on performance with accumulated fatigue and muscle damage. The greater total distance covered and lower RPE in the fifth and 6 LISTs shown by the FIR group were probably due to the FIR lamp therapy effects.

In the present study, the first treatment was provided to the players after the first LIST. The first 90-minute LIST resulted in similar changes in muscle damage and performance markers between the FIR and sham groups (Figure 3), but the changes after the fourth to sixth LISTs were significantly smaller for the FIR than the sham group (Figures 3 and 4). These suggest that overall muscle damage was less for the FIR than sham treatment group, and the FIR treatment facilitated the recovery from LISTs. This supported the hypothesis that a daily FIR lamp therapy over 6 LISTs performed by elite female soccer players would reduce accumulative muscle damage and enhance recovery of performance parameters when compared with a sham treatment. However, it should be noted that the effects of the FIR lamp therapy were not evident in the first 3 or 4 days.

To the best our knowledge, 2 studies have investigated the effect of a FIR treatment (wearing FIR pants for ~8 h during sleep)

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on muscle damage and performance parameters relating to soccer, but they did not find significant effects of the FIR therapy on muscle damage and performance parameters.[11,12] Hausswirth et al[14] showed that a 30-minute FIR lamp therapy applied to the whole body immediately, 1 and 2 days after a 48-minute simulated trail run performed by trained runners improved the recovery of KE MVC torque by 9%; although, no effects on DOMS and plasma CK activity were observed. This was why we used a FIR lamp therapy in the present study, but it should be noted that the FIR lamp therapy used in the present study was different from that in the study by Hausswirth et al[14] in terms of the device and application of the therapy (ie, whole body vs local muscles). The effects of the FIR lamp therapy found in the present study appeared to be greater than those reported by Hausswirth et al[14] such that the recovery of the KE MVC torque was 10% to 12% greater, and peak DOMS was reduced by more than 90% in the FIR group in the present study.

When comparing the effects of the FIR lamp therapy observed in the present study with those by other interventions on soccer players, the effects observed in the present study seem to be greater than those by a CWI,[7] foam rolling,[10] compression garments,[9] and electrical stimulation,[6] although direct comparison is not possible because of the different study designs. For example, Marqués-Jiménez et al[9] examined the effects of compression garments that were worn during and for 3 days (7 h a day) after a friendly soccer match and reported no significant effects on performance and muscle damage markers when compared with a control condition. Rey et al[10] showed that a 20 minute of foam rolling applied to the lower limb muscles at immediately after a soccer training session improved the recovery of agility *T* test (3%) and player's general perception of recovery (18%), and reduced DOMS (40%) at one day later when compared with a control condition, but no significant effects were found on performance parameters. It should be noted that the FIR lamp therapy was repeated every day for 10 days in which the 90-minute LIST was performed on 6 consecutive days in the present study. It is interesting to investigate whether the FIR lamp therapy is effective for enhancing recovery from a single soccer match. It might be that the FIR lamp therapy is more effective for a situation of accumulated fatigue and muscle damage.

The mechanisms underpinning the FIR therapy effects on muscle damage and recovery were not explored in the present study, thus investigating how the FIR lamp therapy worked to attenuate accumulative muscle damage and enhance recovery is the next step. Thus, we could make only some speculations based on in-vivo and in-vitro studies on far-infrared irradiation. It has been documented that FIR produces both nonthermal and thermal effects, including an increase in microvascular dilation, increased blood flow volume, and elevated regional tissue temperature, based on animal studies.[16,17] It seems possible that the FIR lamp treatment effects were due to a combination of increased microvascular blood flow,[17,24] anti-inflammatory effect via the induction of the heme oxygenase-1,[25] increased transforming growth factor-β1 that could increase cellular growth and proliferation, and increased expression myofibroblasts and collagen content on a skin wound area using a rat model,[26] increased nitric oxide, intracellular heat shock proteins, antioxidant effects, and pain relief effects with prostaglandin E_2 production.[15] Future studies are required to examine the mechanisms.

To the best of our knowledge, no previous studies have investigated the prolonged use of FIR lamp radiation on humans. We found only one study that reported a case of a patient who had a deep burn in the ankle as a result of using far-infrared rays (40 min) for the management of chronic ankle pain, but this was due to an inappropriate self-performed treatment.[27] In the present study, we followed the instructions of the use of the FIR lamp such as keeping a distance of 20 cm between the FIR lamp and surface of the targeted muscles and limiting the treatment duration for 30 minutes on each bilateral KE and KF, respectively. Previous studies have shown that a FIR lamp therapy is safe, simple, noninvasive, and painless.[27–30]

Practical Applications

A soccer match induces significant fatigue and muscle damage and reduces neuromuscular function for multiple days after the match, thus strategies to reduce muscle damage and enhance recovery are necessary. The present study found that the FIR lamp therapy was effective for attenuating muscle damage and enhancing recovery from the 90-minute LIST performed over 6 consecutive days, although the therapy required 60 minute every day for 10 days. It appears that the FIR lamp therapy is more effective than the therapeutic modalities that are often used after soccer matches such as CWI and electrical stimulation. Thus, to enhance recovery from a soccer match or a training session, the FIR lamp therapy can be recommended. It is interesting to investigate the effects of the FIR lamp therapy on muscle damage and performance of other team sports athletes, racquet sports players (eg, tennis, badminton), and long-distance runners.

It should be noted that the current study has several limitations. First, the LIST is not necessarily a simulated soccer match but is more focused on football match-related running activities. Thus, the effects of the FIR lamp therapy on actual soccer matches should be investigated in future studies. In the present study, the LIST was performed for 6 consecutive days, which may be rare in practice and does not represent a real soccer tournament schedule. However, it is possible that the present study represented a worst case of muscle damage and fatigue in soccer. Second, the current study did not monitor and control the dietary intake of the participants during the experiment. However, the differences in the measures between the groups did not appear to be explained by the diet. Third, the current study was rather descriptive, and a mechanistic approach to examine the possible mechanisms affecting muscle damage and performance were not investigated. Further studies are warranted to consider these limitations and investigate how the FIR lamp therapy works to affect the body.

Conclusions

The current study showed that the 30-minute daily FIR lamp therapy provided potent effects on reducing muscle damage and enhancing recovery. The underpinning mechanisms of the reduced muscle damage and/or facilitated recovery effects by the FIR lamp therapy warrant further investigations.

Acknowledgments

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